

SUKANTHAN V IIT MADRDAS		ED20B066		
SCHOLASTIC ACHIEVEMENTS				
- Cleared NTSE stage 1 • Qualified for INMO • Certificate of High Distinction in ANCO • JEE Advanced All India Rank: 3430 - Achieved 99.15 Percentile in AIMSCTE • Awarded outstanding student in INSPIRE camp conducted by SSN College				
EDUCATION				
Program		Institute	% / CGPA	Year
Dual degree (B. TECH + M. TECH) Engineering Design, IDDD in Robotics		Indian Institute of Technology, Madras	8.09	2025
Class XII (CBSE)		The PSBB Millennium school OMR	97	2020
Class X (CBSE)		The PSBB Millennium school OMR	96.2	2018
SKILLS				
PROFESSIONAL SKILLS	Languages: Platforms: Embedded platforms: ML/python libraries: Other software: Soft skills:	C/C++, Python , MATLAB, MySQL Windows, Linux, AWS cloud Arduino , Raspberry pi, ESP32 TensorFlow , Keras, OpenCV , Pandas, NumPy, scikit-learn, GPT 3.5, RESTNET50 Mathematica , Figma, Canva, Fusion 360 Effective communication, collaboration, and presentation skills		
PROFESSIONAL EXPERIENCE				
Aprecomm Pvt Ltd. ML Intern May 24 – July 24	Designed and developed AI based Chabot with network intelligence capabilities for customer QoE support. The Chatbot supports customers to solve various issues such as slow internet, no internet, frequent stations disconnect and to monitor the QoE score. Developed it using GPT 3.5 Turbo model with backend cloud interface API for various networking services.			
Siemens Technology Software intern Dec’23 – April 24	Designed and developed a mobile robot live tracking system. The project involves a mobile robot that captures video frames at 30 fps and sends it to the AWS EC2 instance (cloud), which hosts the Kafka broker. Developed python and C++ code for live tracking and video streaming using Apache Kafka . Developed a tracking module to send the data points and developed tracking algorithms such as Lukas Kanade , Back projection .			
Zilogic Systems Software intern June’ 23 – July 23	Developed software for ML based recognition of Seven segment displays from an automobile cluster. Developed an ML model using Random Forest Classifier with sci-kit learn module to detect the seven-segment display numbers (0 to 9). Developed OpenCV based program to identify digit’s position .			
COURSE PROJECTS				
Large Vision model in medical imaging IITM Aug 24-till date	I am currently doing the final year M-Tech project in evaluation of Large Vision models in the field of medical imaging. Developed a RESNET50 based model & TIMM Model to classify the chest X-ray images using transfer learning. Improving the accuracy of ML model using Class Activation Heatmap and Gaze data . This includes the images transformation techniques and preprocessing of the XRAY data from Physion.net.			
Real time road conditioning and Driver Assistance	Developed software for retrofitted device in a car to sense the road conditions, analyze the road condition datapoints in cloud to identify the pothole using ML. - Developed a MQTT subscriber in AWS ECS2 server, a MQTT publisher in the sensor module ESP32). - Developed connection establishment software in IOT device ESP32 to publish the road conditioning events. - Developed the connectivity module for various communication mediums such as Wi-Fi, 5G and Bluetooth for onboarding. - Developed logistic regression model to classify if the given trigger in the accelerometer is a bump or a pothole.			
Obstacle detection	Developed software to find the shortest path andthe shortest distance, from a starting point to a goal point, in the presence of polygonal obstacles using algorithms to detect visibility of a point and Dijkstra algorithm.			
Robotic arm to pickup an object based on color.	- Assembled a robotic arm to lift metallic objects controlled by Arduino. - Welded and calibrated the sensors onto the circuit along with the servo motors. - Used sensors to detect the object and its color by thresholding the intensity of its BGR, using Arduino IDE 2.0.3.			
Courses Taken	- Advanced Machine Learning (Coursera) - Probability, Statistics and Stochastic Process. (MA2040) - Communication Networks for IOT(EE5158) - Digital Signal Processing (ED5017) - Mechanics of Serial Robots (ED6007)			
LEADERSHIP				
CFI Product Design Club Coordinator (May’21- May’22)	- I conducted a workshop to help students design UI for apps using Figma. - Conducted various competitions to help cultivate innovative design thinking. - Collaborated with the electronic club on the Heads-up Display project CFI.			
Avanti Fellowship Mentor (Mar 21-Apr 22)	Trained an underprivileged student for JEE. Provided him with moral support, materials, study plan, exam strategy to join at IIT Delhi, B. Tech.			
Comedy / Oratory club coordinator (Jun 21 – Jun 22)	- Conducted, and hosted various online events for Saarang (Cultural Fest of IITM) - Represented IITM at Inter-IIT, Saarang and Mood Indigo competitions and qualified for various finals			